Project Proposal: Roman Urdu to Urdu Script Conversion

Group Member:

Muhammad Abdullah Awan (2022-SE-08)

---

1. Project Title

Roman Urdu to Urdu Script Conversion using Dictionary-Based and Lightweight ML Approaches

---

2. Motivation

Roman Urdu is widely used on social media, chat applications, and informal communication. However, most computational tools and resources for Urdu NLP are designed for the standard Urdu script. The lack of automatic Roman Urdu to Urdu conversion creates barriers for text processing tasks like sentiment analysis, information retrieval, and machine translation.

This project aims to build a simple yet effective system that converts Roman Urdu into Urdu script, making the text usable for downstream Urdu NLP tasks.

---

3. Research Question

- Can a dictionary-based mapping system reliably convert Roman Urdu to Urdu script?
- How does a lightweight ML-based approach (if applied) compare to rule-based methods?

4. Methodology

Step 1: Data Collection

- Collect small Roman Urdu–Urdu parallel data from:
  - Social media posts (Facebook, Twitter, WhatsApp messages).
  - Existing Roman Urdu corpora (e.g., publicly available transliteration datasets).
  - Create a small dataset manually (few thousand pairs).

Step 2: Preprocessing

- Normalize Roman Urdu spellings (e.g., "kaise", "kesy", "kesay").
- Build a mapping dictionary for common words.

Step 3: Model Development

- **Approach A (Baseline)**: Dictionary-based mapping system → replace Roman Urdu words with Urdu equivalents.
- **Approach B (Optional)**: Lightweight sequence-to-sequence model (LSTM/GRU) if time and GPU resources allow.

Step 4: Evaluation

- **Automatic Metrics**:
  - **Word Accuracy**: Percentage of correctly converted words.
  - **BLEU Score**: N-gram overlap with gold-standard reference.
- **Human Evaluation** (optional):
  - Native speakers rate fluency and correctness of converted text.

5. Expected Outcomes

- A working prototype that converts Roman Urdu text into standard Urdu script.
- A comparative analysis of dictionary-based vs. lightweight ML-based approaches.
- Clear performance metrics (accuracy, BLEU, optional human evaluation).

6. Feasibility & Resource Requirements

- **Computational Resources**: Minimal — dictionary-based system requires no GPU; optional small seq2seq model may use department GPUs.
- **Tools**: Python, NLTK / OpenNMT (optional), UrduHack (for Urdu text processing).
- **Timeframe**:
  - Week 1–2: Data collection & preprocessing.
  - Week 3–4: Dictionary-based system.
  - Week 5–6: Optional ML approach.
  - Week 7: Evaluation & analysis.
  - Week 8: Report writing.

7. Contribution

- A **unique, lightweight Urdu NLP system** for transliteration.
- A **parallel dataset of Roman Urdu–Urdu** (if created, can be shared for future research).
- Evaluation results comparing rule-based and ML-based methods.